DE LA SALLE UNIVERSITY-DASMARINAS
COLLEGE OF INFORMATION AND COMPUTER STUDIES COMPUTER SCIENCE DEPARTMENT

COURSE SYLLABUS
Second Semester, SY 2024-2025

COURSE CODE	:	S-CSTH321
COURSE TITLE	:	THESIS 1
COURSE TYPE	:	Lecture
COURSE CREDIT	:	Two Units
LEARNING MODALITY	:	Flex Blended Learning
PRE-REQUISITES	:	Should be 3rd year level
CO-REQUISITES	:	None
PROFESSOR	:	Ms. Josephine T. Eduardo
CONSULTATION HOURS	:	H/1300-1600/COS100-C
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VISION

A leading Catholic university that inspires excellence and drives innovation toward a just, peaceful, and sustainable society.

MISSION

To champion the Human and Christian Education of lifelong learners who value history and culture through responsive and inclusive academic, research, and extension programs.

COURSE DESCRIPTION:

This course improves your understanding of the various methods of research, some basic statistical designs and procedures, and the different parts of a thesis, including field and laboratory performance. The course output is an undergraduate thesis consisting of the defended proposal, results and discussions, and conclusion. This course includes the practical applications of Godly stewardship of nature through proper handling of different test organisms and research settings. It also includes character-building through proper experimentation and scientific paper oral presentation.

COURSE LEARNING OUTCOMES:

By the end of this course, students can:

CLO1. understand the latest concepts in systems design, software engineering and current technologies;
CLO2. investigate and evaluate prominent literature connected to the research topic;
CLO3. define the scope of the project, planning and scheduling the collaborative efforts of the team, and documenting the effort via oral and written proposals;
CLO4. demonstrate that they can function, communicate, collaborate and continue to learn effectively as ethically and socially responsible CS professionals;
CLO5. develop skills to communicate with and balance the interests of multiple stakeholders (both technical and non-technical) on a project; and
CLO6. present to the assigned panelists the software for applications (Mobile apps, Game Creation, Artificial Intelligence, Internet of Things, Recommender Systems, Computer Vision, Natural Language Processing, Reinforcement Learning, Deep Learning and Machine Learning) with the necessary documentation.

LEARNING PLAN:

Learning Outcomes for Midterm Period

Course Learning Outcomes	Topic Learning Outcomes
CLO1. Understand the latest concepts in system design, software engineering and current technologies.	TLO1. Apply concepts of computing and mathematics in different domains of computer science. TLO2. Identify and analyze problems and define computing requirements appropriate to their solutions.
CLO2. Investigate and evaluate prominent literature connected to the research topic.	TLO3. Design, develop, test and evaluate various solutions using algorithms for complex computing problems that take full advantage of usability of computer resources.
CLO3. Define the scope of the project, planning and scheduling the collaborative efforts of the team, and documenting the effort via oral and written proposals.	TLO4. Recognize the legal, social ethical, and professional issues involved in the utilization of computer technology and address these issues.

Module No and Learning Outcomes	Week	Teaching-Learning Activities / Assessment Strategies	Technology Enabler	Onsite / F2F	Asynchronous	Allotted Hours
Module 0 CLO1 TLO1	Week 1 Jan 20-25	• Gospel Reading, Reflection, and Prayer • Discussion of Course Syllabus • Discussion of course policies and requirements • Discussion of CICS research manual and thesis project guidelines **Module 0: Course Introduction**	Schoolbook	✓		3
Module 1 CLO1 TLO1 TLO2	Week 2 Jan 27-Feb 1	• Gospel Reading, Reflection, and Prayer • Discussion of Research Introduction • Discussion of classification and characteristic of research • Thesis groupings **Module 1: Introduction to Research**	Schoolbook	✓		3
	Week 3 Feb 3 - 8	• Gospel Reading, Reflection, and Prayer • Enabling Assessment – Short Quiz in Module 1	Schoolbook		✓	3

Module 2 CLO1 CLO2 TLO1 TLO2 TLO3	Week 4 Feb 10-15	• Gospel Reading, Reflection, and Prayer • Discussion of the following: - Background of the study - The research problem - Statement of Objectives - Significance of the study - Scope and limitations of the study **Module 2: Chapter 1 Introduction**	Schoolbook	✓		3
	Week 5 Feb 17-22	• Gospel Reading, Reflection, and Prayer • Endorsement to Research Adviser • Enabling Assessment – Identify and explore three potential research topic	Schoolbook		✓	3
	Week 6 Feb 24-Mar 1	• Gospel Reading, Reflection, and Prayer • Enabling Assessment - Submissions of three research topic proposals	Schoolbook		✓	3
Module 3 CLO1 CLO2 CLO3 TLO1 TLO2 TLO3	Week 7 Mar 3-8	• Gospel Reading, Reflection, and Prayer • Discussion of the following: - Related Literature and Studies - References and Citations **Module 3: Chapter 2 Review of Related Literature**	Schoolbook	✓		3

Module 4 CLO3 CLO4 TLO3 TLO4	Week 8 Mar 10-15	• Gospel Reading, Reflection, and Prayer • **Sustainability Week** • Discussion of ethical issues in research and intellectual property rights **Module 4: Research Ethics and Intellectual Rights**			✓	3
Midterm Exam	Week 9 Mar 17-22	• Gospel Reading, Reflection, and Prayer • **Midterm Summative Assessment on Module 1 - 4**		✓		3
TOTAL				15	12	27

Learning Outcomes for Final Period

Course Learning Outcomes	Topic Learning Outcomes
CLO4. Demonstrate that they can function, communicate, collaborate, and continue to learn effectively as ethically and socially responsible Computer Science professionals.	TLO5. Document computer-based systems by writing effective reports of risk assessment impact.
CLO5. Develop skills to communicate with and balance the interest of multiple stakeholders (both technical and non-technical) on project.	TLO6. Function effectively as a member or leader of a development team recognizing the different roles with a team to accomplish a common goal.
CLO6.Present to the assigned panelists the software applications (Mobile apps, Game, Artificial Intelligence, Internet of Things, Recommender System, Computer Vision, Natural Language Processing, Reinforcement Learning, Deep Learning and Machine Learning) with the necessary documentation.	TLO7. Engage in continuing professional development through performing independent or collaborative research as computer professionals. TLO8. Work effectively as an individual or as a member of a team during software development.

Module No and Learning Outcomes	Week	Teaching-Learning Activities / Assessment Strategies	Technology Enabler	Onsite / F2F	Asynchronous	Allotted Hours
Module 5 CLO4 TLO5	Week 10 Mar 24-29	• Gospel Reading, Reflection, and Prayer • Discussion of Research Methods **Module 5: Research Methodology**	Schoolbook	✓		3
Research Work	Week 11 Mar 31-Apr 5	• Gospel Reading, Reflection, and Prayer • Consultation with adviser • Documentation and software development	Schoolbook		✓	3
Research Work	Week 12 Apr 7-12	• Gospel Reading, Reflection, and Prayer • Consultation with adviser • Documentation and software development • Enabling Assessment – Submission of Progress Report	Schoolbook		✓	3
	Week 13 Apr 14-19	***Independent Learning Days (April 14-15)*** ***Holy Week (April 16-19)***			✓	3

	Week 14 Apr 21-26	***Research Week***			✓	3
Research Work	Week 15 Apr 28-May 3	• Gospel Reading, Reflection, and Prayer • Consultation with adviser • Documentation and software development	Schoolbook		✓	3
Research Work	Week 16 May 5-10	• Gospel Reading, Reflection, and Prayer • Consultation with adviser • Documentation and software development • Endorsement of Proposal Defense	Schoolbook		✓	3
Proposal Defense	Week 17 May 12-17	• Gospel Reading, Reflection, and Prayer • Proposal Defense (Finals Summative Assessment)		✓		2
Final Exam	Week 18 May 19-24	• Gospel Reading, Reflection, and Prayer • Proposal Defense (Finals Summative Assessment)		✓		3
TOTAL				9	18	27

GRADING SYSTEM:

Midterm	**%**	**Final**	**%**

Enabling Assessments	50%	Enabling Assessments	50%
Summative Assessment	30%	Summative Assessment Proposal Defense Grade (60% Panelists; 40% Research Adviser)	30%
Class Participation	20%	Class Participation	20%
	100%		100%

FINAL GRADE = $\frac{\text{Midterm Grade + Final Grade}}{2}$

COURSE POLICIES AND REQUIREMENTS:

1. This course follows the Flex Blended Learning Mode.

2. Enrollment in an E-Class. The Center for Innovative Learning Programs (CILP) shall enroll you in the e-class. This will be based on the class list provided by the Office of the University Registrar and at the portal.

3. Communication and Feedback
    a. Onsite classes shall happen as indicated in the learning plan section of the syllabus. Fifty percent (50%) of the weekly classes is delivered face-to-face while the remaining is asynchronous.
    b. Online Asynchronous Communication shall happen when the faculty utilizes at least one of the recommended asynchronous communications. The minimum response time for asynchronous communication is 48 hours, excluding Sundays and holidays. All communication platforms must be explicitly indicated in the syllabus. Below are the recommended asynchronous communication platforms:
        i. Forum – at the Forum Tab at Schoolbook, students can post their general concerns. The forum is public, and any post can be seen by the class.
        ii. Messaging – at Schoolbook. Any private or confidential matters may be messaged through the Schoolbook messaging feature.
        iii. Email – you may email your professor. (Professor's email will be given during the course orientation)
    c. No use of foul, curse, and other bad words, whether written or verbal. No shouting. No flooding of chat and messages. No selling.

4. Attendance Policy
    a. Attendance Policy for Onsite Classes is scheduled based on the learning plan. A maximum of 20% for class participation/recitation/attendance shall be allotted for the student's grade.
    b. The onsite meeting will only be held at the schedule indicated in the registration form.
    c. Students are required to attend onsite classes, 40% of absence (from onsite classes) for the entire semester means 0.0 final grade.

5. Intellectual Property Policy
    a. Policy and copyright infringement will not be tolerated. Any violation of such will earn students a grade of zero in the assessment.
    b. Using any illegally obtained software is strictly prohibited.
    c. Students are encouraged to use their original photos, videos, and other resources. Otherwise, students can use royalty-free resources or embed the sources in their submissions to avoid copyright infringement and/or plagiarism.
    d. Giving passwords to Schoolbook and Office 365 is strictly prohibited. Likewise, accessing Schoolbook and Office 365 accounts other than the students' personal account is also strictly prohibited. Violating students will be reported to the Student Welfare and Formation Office (SWAFO).

6. Other Policies
    a. Attire - students must wear decent casual or smart-casual attire prescribed by the Student Handbook.
    b. Students must be honest at all times; cheating and plagiarism in any form will merit a grade of 0.00.

c. Cellular/Mobile phones should always be in silent mode during class hours; the use of cellular phones is prohibited in class unless special permission is sought. Cellular phones cannot also be used as calculators during examinations.

d. Any complaints against the teacher or against classmates should be properly addressed to the subject teacher for appropriate action. Students may seek the help and guidance of their academic/registration adviser in resolving the issue with the subject teacher when such mediation is needed.

REFERENCES:

Online References

Call Number/ E-provider	Reference Material
Online	Naoum, S. 2019. Dissertation research and writing for built environment students. London: Taylor & Francis; [cited 2024 Jan 23]. Available from: https://www.taylorfrancis.com/books/mono/10.1201/9781351203913/dissertation-research-writing-built-environment-students-shamil-naoum
Journal	**Rosas MF, Ambat SC, Lagman AC.** 2019. Extraction of students' feedback in university services using logistic regression and chi-square automatic interaction detection (CHAID) algorithm. *Int J Simul Syst Sci Technol*. Apr 2019;20(2):1-8. Available from: https://ijssst.info/Vol-20/No-2/paper8.pdf **Eduardo JT, Barrameda RB, Rosas MF, Lagman AC, Ballera M.** 2019. Applying cellular automata in finding vascular plants domain Josephine. *Int J Simul Syst Sci Technol*. 2019 Supplement 2;20:9.1-9.7. Available from: https://ijssst.info/Vol-20/NoS2/paper9.pdf Herrera ALV, Tecson PDI, Eduardo JT, Rosas MF, Barrameda RB. 2019. Utilizing Simultaneous Localization and Mapping (SLAM) in Augmented-Reality Shell Game for iOS. *Test Eng Manag*. 81:4973-4979. The Mattingley Publishing Co., Inc. Eduardo JT, Rosas MF, Mayuga ED, Barrameda RB. 2019. Fuzzy Logic: A Technique for Assessing Students' Learning Performance. *Test Eng Manag*. 81:5213-5217. The Mattingley Publishing Co., Inc. Lalas GL, Canto KA, Uy GCN, Rosas MF, Eduardo JT. 2019. SaveYourGrades! A 3D Mobile Educational Entertainment Game Using Ant Colony Algorithm. *Test Eng Manag*. 81:5218-5225. The Mattingley Publishing Co., Inc. Alday RP, Rosas MF. 2019. Business Intelligence Solution for Bikers Haven Restaurant. In: *2019 IEEE 10th Annual Ubiquitous Computing, Electronics & Mobile Communication Conference (UEMCON)*, New York City, NY, USA; 2019. p. 1204-1210. doi:10.1109/UEMCON47517.2019.8992956. Available from: https://ieeexplore.ieee.org/abstract/document/8992956 Galarosa JE, Encarnacion MM, Rosas MF, Gatpandan P. 2019. Predicting the Relationship between Academy of Dasmarinas using a Correlation and Clustering Analysis. *Test Eng Manag*. 81:5226-5235. The Mattingley Publishing Co., Inc.

Prepared: **Computer Science Department**
2nd Semester 2024-2025



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